

MATH3026/4022 Lecture 6

The Partial Autocorrelation Function and Sample Statistics

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- ▶ Definition of the partial autocorrelation function (PACF)
- ▶ Calculation of the PACF
- ▶ Sample statistics from observed time series data
- ▶ Bartlett's formula

Partial Autocorrelation Function (PACF)

We have seen that the autocorrelation function for a time series, $\rho(h)$, is defined as the correlation between X_t and X_{t+h} (i.e. at lag h). That is

$$\rho(h) = \text{Cor}(X_t, X_{t+h}).$$

For example, for a (weakly) stationary $AR(1)$ process

$$X_t = \phi X_{t-1} + Z_t$$

with $|\phi| < 1$, the ACF at lag h is

$$\rho(h) = \phi^{|h|}$$

(see Lecture 3).

Partial Autocorrelation Function (PACF)

For the $AR(1)$ process, we know that

$$\begin{aligned}X_t &= \phi X_{t-1} + Z_t; \\X_{t-1} &= \phi X_{t-2} + Z_{t-1}.\end{aligned}$$

So we see that there is dependence between X_t and X_{t-1} and also dependence between X_{t-1} and X_{t-2} .

A natural question might be 'Could the dependence between X_t and X_{t-2} be explained by the mutual dependence of X_t and X_{t-1} on X_{t-1} ?'

To answer a question like this, we introduce the **partial autocorrelation function** (PACF).

Partial Autocorrelation Function (PACF)

Definition 6.1: Partial Autocorrelation Function (PACF)

The partial autocorrelation function (PACF) between X_t and X_{t-h} is a measure of the correlation between X_t and X_{t-h} with the effect of $X_{t-1}, X_{t-2}, \dots, X_{t-h+1}$ removed.

Now we must consider how to calculate the PACF. To do this we shall introduce some new notation.

Calculation of the PACF

We assume that the time series $\{X_t: t \in \mathbb{Z}\}$ is a (weakly) stationary process with mean zero.

To simplify the notation, we define $\gamma(h) = \gamma_h$ and $\rho(h) = \rho_h$.

The vector $\mathbf{X}_m$ and the matrix Γ_m are defined

$$\mathbf{X}_m = \begin{bmatrix} X_{t-1} \\ X_{t-2} \\ \vdots \\ X_{t-m} \end{bmatrix}$$

$$\Gamma_m = \mathbb{E} \left(\mathbf{X}_m \mathbf{X}_m^\top \right) = \begin{bmatrix} \gamma_0 & \gamma_1 & \gamma_2 & \cdots & \gamma_{m-1} \\ \gamma_1 & \gamma_0 & & & \\ \gamma_2 & & \gamma_0 & & \\ \vdots & & & \ddots & \\ \gamma_{m-1} & & & & \gamma_0 \end{bmatrix}.$$

Calculation of the PACF

We define the vector of covariances γ_m as

$$\gamma_m = \mathbb{E}[X_t \mathbf{X}_m] = \begin{bmatrix} \mathbb{E}(X_t X_{t-1}) \\ \mathbb{E}(X_t X_{t-2}) \\ \vdots \\ \mathbb{E}(X_t X_{t-m}) \end{bmatrix} = \begin{bmatrix} \gamma_1 \\ \gamma_2 \\ \vdots \\ \gamma_m \end{bmatrix}.$$

Now we consider a regression model where we regress X_t on $X_{t-1}, \dots, X_{t-m}$, defined as

$$\begin{aligned} X_t &= \beta_{m,1} X_{t-1} + \beta_{m,2} X_{t-2} + \dots + \beta_{m,m} X_{t-m} + \epsilon_t \\ &= \mathbf{X}_m^\top \boldsymbol{\beta}_m + \epsilon_t \end{aligned} \tag{6.1}$$

where ϵ_t is a mean zero error term with a constant variance and ϵ_t is uncorrelated with $X_{t-1}, \dots, X_{t-m}$.

Calculation of the PACF

As with other linear regression models that you may have seen, the parameters

$$\boldsymbol{\beta}_m = \begin{bmatrix} \beta_{m,1} \\ \beta_{m,2} \\ \vdots \\ \beta_{m,m} \end{bmatrix}$$

can be estimated using the least squares method.

That is, by minimising the sum of squared distances, D , where

$$D = \sum_t (X_t - \mathbf{X}_m^\top \boldsymbol{\beta}_m)^2.$$

Calculation of the PACF

If we multiply both sides of (6.1) by X_{t-m} and take expectations, we obtain

$$\begin{aligned}\mathbb{E}(X_{t-m}X_t) &= \mathbb{E}(X_{t-m}\mathbf{X}_m^\top\boldsymbol{\beta}_m) + \mathbb{E}(X_{t-m}\epsilon_t) \\ &= \sum_{i=1}^m \beta_{m,i}\mathbb{E}(X_{t-i}X_{t-m}) + 0 \quad (X_{t-m}, \epsilon_t \text{ uncorrelated}) \\ &= \sum_{i=1}^m \beta_{m,i}\mathbb{E}(X_{t-i}X_{t-m}).\end{aligned}$$

Since $\mathbb{E}(X_{t-m}X_t) = \gamma_m$ and $\mathbb{E}(X_{t-i}X_{t-m}) = \gamma_{m-i}$, we can rewrite the above as

$$\gamma_m = \sum_{i=1}^m \beta_{m,i}\gamma_{m-i}.$$

Calculation of the PACF

Looking at the expression

$$\gamma_m = \sum_{i=1}^m \beta_{m,i} \gamma_{m-i}.$$

we see that this is the autocovariance at lag m written as a linear combination of the autocovariances at lags $m-1, \dots, 1$ plus a part that depends on the variance of the process (γ_0).

We can write this as

$$\gamma_m = \underbrace{\beta_{m,1}\gamma_{m-1} + \dots + \beta_{m,m-1}\gamma_1}_{\text{Dependence on autocovariances at intermediate lags}} + \beta_{m,m}\gamma_0$$

The parameter $\beta_{m,m}$ reflects the contribution to the autocovariance at lag m having accounted for autocovariances at lags $1, \dots, m-1$.

Calculation of the PACF

Using the vector/matrix notation defined earlier, we see that

$$\gamma_m = \Gamma_m \beta_m.$$

Solving for β_m , we obtain

$$\beta_m = \Gamma_m^{-1} \gamma_m.$$

Now, we would like to determine the partial autocorrelation function but the above expressions are written in terms of autocovariances.

Suppose that we define

$$\rho_m = \frac{\gamma_m}{\gamma_0}, \quad P_m = \frac{1}{\gamma_0} \Gamma_m \quad \text{and} \quad \boldsymbol{\rho}_m = \frac{1}{\gamma_0} \gamma_m.$$

Calculation of the PACF

Then

$$\begin{aligned}\boldsymbol{\rho}_m &= \frac{1}{\gamma_0} \boldsymbol{\gamma}_m \\ &= \frac{1}{\gamma_0} \boldsymbol{\Gamma}_m \boldsymbol{\beta}_m \\ &= \mathbf{P}_m \boldsymbol{\beta}_m.\end{aligned}$$

Hence

$$\boldsymbol{\beta}_m = \mathbf{P}_m^{-1} \boldsymbol{\rho}_m$$

where, as before,

$$\boldsymbol{\beta}_m = \begin{bmatrix} \beta_{m,1} \\ \beta_{m,2} \\ \vdots \\ \beta_{m,m} \end{bmatrix}.$$

Calculation of the PACF

Here, $\beta_{m,m}$ is the final element of β_m and we note that $\beta_{m,m}$ provides a measure of relationship between X_t and X_{t-m} , having accounted for $X_{t-1}, \dots, X_{t-m+1}$.

In fact, $\beta_{m,m}$ is the **partial autocorrelation** between X_t and X_{t-m} (i.e. the correlation between X_t and X_{t-m} with the effect of $X_{t-1}, \dots, X_{t-m+1}$ removed).

As a simple example, where $m = 1$

$$\begin{aligned}\beta_{1,1} &= P^{-1}\rho_1 \\ &= \rho_0\rho_1 \quad (\text{since } P \text{ is the } 1 \times 1 \text{ matrix } [\rho_0]) \\ &= \rho_1 \quad (\text{since } \rho_0 = 1).\end{aligned}$$

Calculation of the PACF: $m = 2$

Calculation of the PACF

In general, to determine the PACF at lag m ($\beta_{m,m}$), we solve

$$\begin{bmatrix} 1 & \rho_1 & \rho_2 & \cdots & \rho_{m-1} \\ \rho_1 & 1 & & & \\ \rho_2 & & 1 & & \\ \vdots & & & \ddots & \\ \rho_{m-1} & & & \rho_1 & 1 \end{bmatrix} \begin{bmatrix} \beta_{m,1} \\ \beta_{m,2} \\ \vdots \\ \beta_{m,m} \end{bmatrix} = \begin{bmatrix} \rho_1 \\ \rho_2 \\ \vdots \\ \rho_m \end{bmatrix}.$$

We need to solve a system of m linear equations with m unknowns, written

$$\mathbf{P}_m \boldsymbol{\beta}_m = \boldsymbol{\rho}_m.$$

Assuming that the determinant of $\mathbf{P}_m$ is non-zero, we can use **Cramer's rule** to solve for $\beta_{m,m}$.

Definition 6.2: Cramer's Rule

Suppose that A is an $n \times n$ matrix with a non-zero determinant and that

$$A\mathbf{x} = \mathbf{b}$$

is a system of linear equations with unknowns $\mathbf{x} = (x_1, \dots, x_n)^\top$ (here $\mathbf{b} \in \mathbb{R}^n$). Then

$$x_i = \frac{\det A_i}{\det A} \quad \text{for } i = 1, \dots, n.$$

where A_i is the matrix A but where the i^{th} column of A has been replaced by $\mathbf{b}$.

Cramer's Rule

Using Cramer's rule, we see that the PACF at lag m can be evaluated as

$$\beta_{m,m} = \frac{\det P_m^*}{\det P_m}$$

where P_m^* is the matrix P_m in which the m^{th} column has been replaced by ρ_m .

Recursive Calculation of the PACF

We can also calculate the PACF using a recursive method, which can be useful in some situations.

Here, we introduce the notation

$$\begin{aligned}a_1, a_2, \dots, a_m &= \beta_{m,1}, \beta_{m,2}, \dots, \beta_{m,m}; \\b_1, b_2, \dots, b_{m+1} &= \beta_{m+1,1}, \dots, \beta_{m+1,m+1}.\end{aligned}$$

When $m = 2$, we see that

$$\begin{aligned}\begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} &= \begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix}.\end{aligned}$$

Recursive Calculation of the PACF

We obtain the pair of simultaneous equations

$$\rho_1 = a_1 + a_2\rho_1; \quad (6.2a)$$

$$\rho_2 = a_1\rho_1 + a_2. \quad (6.2b)$$

In a similar manner

$$\rho_3 = b_1\rho_2 + b_2\rho_1 + b_3; \quad (6.3a)$$

$$\rho_2 = b_1\rho_1 + b_2 + b_3\rho_1; \quad (6.3b)$$

$$\rho_1 = b_1 + b_2\rho_1 + b_3\rho_2. \quad (6.3c)$$

Rearranging (6.3c) and (6.3b) we see that

$$b_1 + b_2\rho_1 = \rho_1 - b_3\rho_2; \quad (6.4)$$

$$b_1\rho_1 + b_2 = \rho_2 - b_3\rho_1.$$

Recursive Calculation of the PACF

Then (6.4) can be written

$$\begin{aligned}\begin{bmatrix} 1 & \rho_1 \\ \rho_1 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} &= \begin{bmatrix} \rho_1 - b_3 \rho_2 \\ \rho_2 - b_3 \rho_1 \end{bmatrix} \\ \Rightarrow P_2 \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} &= \begin{bmatrix} \rho_1 - b_3 \rho_2 \\ \rho_2 - b_3 \rho_1 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} &= P_2^{-1} \begin{bmatrix} \rho_1 - b_3 \rho_2 \\ \rho_2 - b_3 \rho_1 \end{bmatrix} \\ &= P_2^{-1} \begin{bmatrix} \rho_1 \\ \rho_2 \end{bmatrix} - b_3 P_2^{-1} \begin{bmatrix} \rho_2 \\ \rho_1 \end{bmatrix} \\ &= \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} - b_3 \begin{bmatrix} a_2 \\ a_1 \end{bmatrix} .\end{aligned}$$

Recursive Calculation of the PACF

We see that

$$\begin{aligned}b_1 &= a_1 - a_2 b_3 \\b_2 &= a_2 - a_1 b_3\end{aligned}\tag{6.5}$$

Recall (6.3a):

$$\rho_3 = b_1 \rho_2 + b_2 \rho_1 + b_3.$$

Substitution of (6.5) into (6.3a) yields

$$\begin{aligned}\rho_3 &= \rho_2(a_1 - b_3 a_2) + \rho_1(a_2 - b_3 a_1) + b_3 \\&= b_3(1 - a_1 \rho_1 - a_2 \rho_2) + a_1 \rho_2 + a_2 \rho_1\end{aligned}$$

Rearranging, we see that

$$b_3 = \beta_{3,3} = \frac{\rho_3 - a_1 \rho_2 - a_2 \rho_1}{1 - a_1 \rho_1 - a_2 \rho_2}.$$

Recursive Calculation of the PACF

In general, the PACF may be written down as follows

$$\begin{aligned}\rho(0, m+1|1, \dots, m) &= \beta_{m+1, m+1} = b_{m+1} \\ &= \frac{\rho_{m+1} - a_1 \rho_m - \dots - a_m \rho_1}{1 - a_1 \rho_1 - \dots - a_m \rho_m};\end{aligned}$$

with

$$b_j = a_j - b_{m+1} a_{m+1-j} \quad \text{for } j = 1, \dots, m.$$

We note that this method is sensitive to rounding errors, when using estimates, especially when parameters are close to the boundary of the admissible region for stationarity.

Example 6.1

Consider the $AR(1)$ process

$$X_t = \phi X_{t-1} + Z_t$$

where $|\phi| < 1$. Calculate the PACF for this process.

Example 6.1

PACF - $AR(p)$ process

In general, for the $AR(p)$ process, the PACF cuts off to zero **after** lag p .

Why? Recall Equation (6.1) (written below) when we derived the PACF:

$$X_t = \beta_{m,1}X_{t-1} + \beta_{m,2}X_{t-2} + \dots + \beta_{m,m}X_{t-m} + \epsilon_t$$

The above is an autoregressive model for X_t and $\beta_{m,m}$ (the coefficient of X_{t-m}) is the PACF at lag m .

For an $AR(p)$ process, we know that $\beta_{m,m} = 0$ where $m > p$ and, hence, the PACF is zero for all $m > p$.

This property of the PACF for an $AR(p)$ may help us to identify whether or not sampled data are from an AR process – more on this during Lecture 7.

Example 6.2

Consider the $AR(2)$ process

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t$$

where $|\phi_1| < 1$ and $|\phi_2| < 1$. Calculate the PACF for this process.

Example 6.2

Example 6.3

Suppose that $\{X_t\}$ is an $MA(1)$ process, written

$$X_t = \theta Z_{t-1} + Z_t$$

Derive the PACF for $\{X_t\}$.

Example 6.3

Example 6.3

Time Series: Sample Statistics

So far, we have explored various theoretical properties of time series including stationarity, autocovariance, autocorrelation and partial autocorrelation in the context of common time series models (e.g. $AR(p)$, $MA(q)$ etc.).

Now, suppose we have some observed data from a time series and we would like to estimate these measures (perhaps when choosing the type of model that we'd like to fit to the observed data).

We will now turn our attention to sample statistics for observed time series data and how these statistics may be of help when fitting models.

Time Series: Sample Statistics

Suppose that $\mathbf{x}_n = (x_1, \dots, x_n)^\top$ is a set of n observations from a process/time series $\{X_t\}$, taken at successive times.

We shall assume that $\{X_t\}$ is stationary with constant mean μ .

Note from Lecture 2 that if a time series does not appear to be stationary (e.g. when plotting $\mathbf{x}_n$ against time), we might use an operator (e.g. B) to transform the series into a stationary time series.

We assume that the time series (from which the observed data originate) can be described by the following underlying model

$$\phi(B)(X_t - \mu) = \theta(B)Z_t.$$

Time Series: Sample Statistics

Since the time series is stationary, the mean, μ , may be estimated using the sample mean as

$$\hat{\mu} = \bar{x} = \frac{1}{n} \sum_{t=1}^n x_t.$$

The sample covariance between X_t and X_{t+h} (i.e. at lag h) is defined

$$c_h = \frac{1}{n} \sum_{t=1}^{n-h} (x_t - \bar{x})(x_{t+h} - \bar{x})$$

Notably, c_0 (the estimated variance of the process) is

$$c_0 = \frac{1}{n} \sum_{t=1}^n (x_t - \bar{x})^2.$$

We note that c_h is biased for $\gamma(h)$ but that the bias is of order $1/n$ (i.e. small where n is large).

Time Series: Sample Statistics

We might wonder why the divisor n is used in the formula for c_h and not $n - h - 1$?

For example, we know that the sample variance is an unbiased estimator for the population variance when we divide $\sum_{t=1}^n (x_t - \bar{x})^2$ by $n - 1$.

In reality, using the divisor n when computing c_h results in some useful properties when c_h is used to estimate the **spectrum** of a time series (to be covered in a later lecture).

The autocorrelation function at lag h is estimated by

$$r_h = \frac{c_h}{c_0} = \frac{\sum_{t=1}^{n-h} (x_t - \bar{x})(x_{t+h} - \bar{x})}{\sum_{t=1}^n (x_t - \bar{x})^2}.$$

Finally, we consider estimation of the partial autocorrelation.

Recall that the partial autocorrelation at lag m is the term $\beta_{m,m}$ where $\beta_{m,m}$ is the m^{th} element of $P_m^{-1} \boldsymbol{\rho}_m$.

A natural estimator for the PACF might be obtained by substituting the matrix P_m with a matrix that contains **sample** autocorrelations.

Time Series: Sample Statistics

That is, define the matrix R_m and the vector $\mathbf{r}_m$ where

$$R_m = \begin{bmatrix} 1 & r_1 & r_2 & \dots & r_{m-1} \\ r_1 & 1 & & & \\ r_2 & & 1 & & \\ \vdots & & & \ddots & r_1 \\ r_{m-1} & & & r_1 & 1 \end{bmatrix}, \quad \mathbf{r}_m = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}.$$

Then the estimate of the PACF at lag m is given by $\hat{\beta}_{m,m}$ where

$$\hat{\beta}_{m,m} = m^{\text{th}} \text{ element of } R_m^{-1} \mathbf{r}_m.$$

Time Series: Sample Statistics

An alternative to approach to estimating the PACF could be based on the recursive PACF calculation method, seen earlier.

With this method, we saw that

$$\rho(0, m+1|1, \dots, m) = \beta_{m+1, m+1} = \frac{\rho_{m+1} - a_1\rho_m - \dots - a_m\rho_1}{1 - a_1\rho_1 - \dots - a_m\rho_m};$$

with

$$b_j = a_j - b_{m+1}a_{m+1-j} \quad \text{for } j = 1, \dots, m.$$

To obtain the sample equivalent, we might replace each of $\rho_1, \dots, \rho_m$ with $r_1, \dots, r_m$ yielding the estimate

$$\hat{\beta}_{m+1, m+1} = \frac{r_{m+1} - a_1r_m - \dots - a_mr_1}{1 - a_1r_1 - \dots - a_mr_m}.$$

Bartlett's Formula

Bartlett's formula for $\text{Var}(r_j)$ defines a useful relationship between the sample ACF and the theoretical ACF for a time series and is defined

$$\text{Var}(r_j) \approx \frac{1}{n} \sum_{i=-\infty}^{\infty} (\rho_i^2 + \rho_{i-j}\rho_{i+j} - 4\rho_i\rho_j\rho_{i+j} + 2\rho_i^2\rho_j^2).$$

In addition

$$\text{Corr}(r_h, r_{h+k}) \approx \frac{\sum_{j=-\infty}^{\infty} \rho_j \rho_{j+k}}{\sum_{j=-\infty}^{\infty} \rho_j^2}.$$

The correlation between r_h and r_{h+k} can be large, indicating that, in some cases, estimates r_h, r_{h+k} of ρ_h, ρ_{h+k} may be highly dependent.

Bartlett's Formula

Bartlett's formula may be useful in identifying properties of time series. For example:

If the underlying process is $MA(q)$ then we know that $\rho_{q+1} = \rho_{q+2} = \dots = 0$ and then, for $j > q$

$$\text{Var}(r_j) \approx \frac{1}{n} \left[\rho_0^2 + 2 \sum_{i=1}^q \rho_i^2 \right] = \frac{1}{n} \left[1 + 2 \sum_{i=1}^q \rho_i^2 \right].$$

If the underlying process is white noise then we know that $\rho_i = 0$ for $i \neq 0$ and then

$$\text{Var}(r_j) \approx \frac{1}{n} \quad \text{for } j = 1, 2, \dots$$

Bartlett's Formula

Moreover, suppose that the underlying process is white noise and we assume that r_j is normally distributed with mean zero and variance $\frac{1}{n}$. That is

$$r_j \sim \mathcal{N}\left(0, \frac{1}{n}\right)$$

Then, standardising, we see that

$$r_j\sqrt{n} \sim \mathcal{N}(0, 1)$$

and we can use the following probability statement to make inference about ρ_j

$$\mathbb{P}(-1.96 < r_j\sqrt{n} < 1.96) = 0.95.$$

Bartlett's Formula

Then

$$\mathbb{P} \left(-\frac{1.96}{\sqrt{n}} < r_j < \frac{1.96}{\sqrt{n}} \right) = 0.95$$

In words, we might expect r_j to lie in the above interval for most j .

As such, if we think that a process is white noise with mean zero, we could plot the sample autocorrelations, r_j against j and we should observe that most of the sample autocorrelations should lie in the interval

$$\left(-\frac{2}{\sqrt{n}}, \frac{2}{\sqrt{n}} \right)$$

Here, we have rounded 1.96 to 2 to be conservative.

Lecture 6: Learning Outcomes

- ▶ Understand what is meant by the partial autocorrelation function (PACF) and know how to calculate the PACF.
- ▶ Describe and use sample statistics for observed time series data
 - ▶ Sample mean
 - ▶ Sample covariance/correlation
 - ▶ PACF estimation
- ▶ Bartlett's formula and its use in identifying properties of observed time series.